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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

4884117682

MATHEMATICS (SYLLABUS D) **4024/13**

Paper 1 **October/November 2010**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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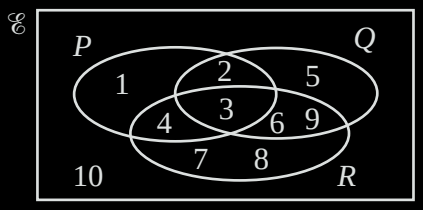
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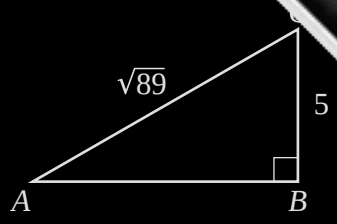


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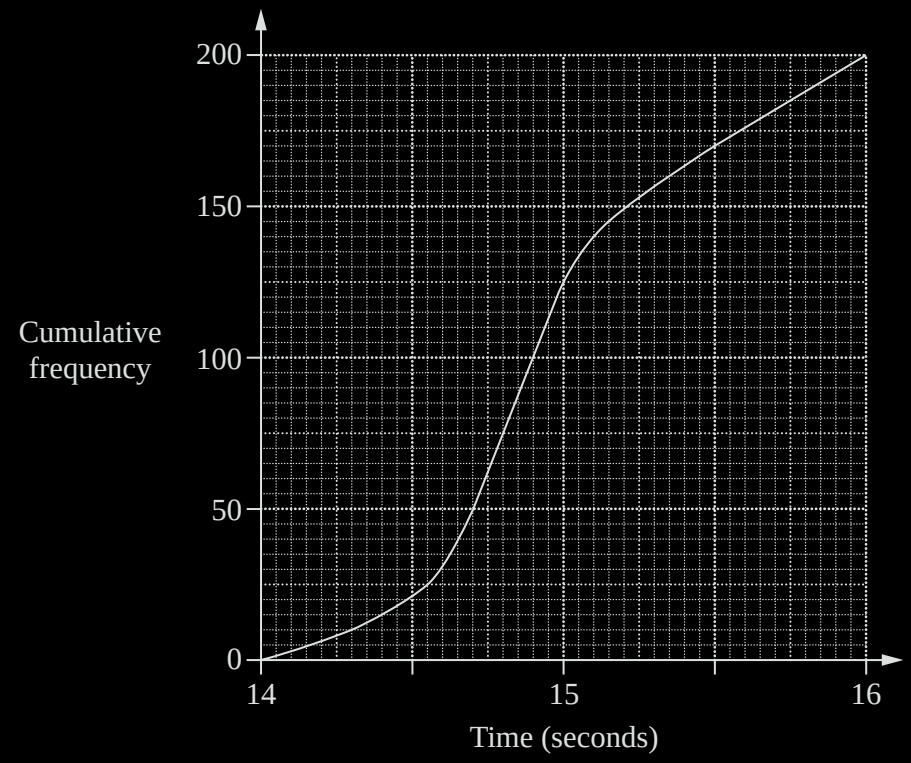


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Use the curve to find

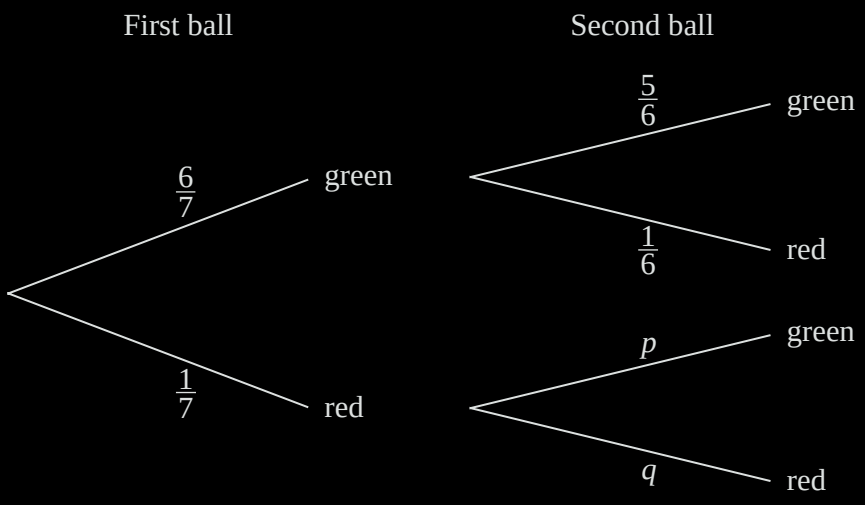
(a) the lower quartile,

Answer (*a*) ..... seconds [1]

(b) the number of children who took at least 15.5 seconds.

Answer (*b*) .....[2]

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(a) Find the values of p and q.

Answer (*a*) p = ... q = ...[1]

(b) Expressing each answer as a fraction in its simplest form, find the probability that

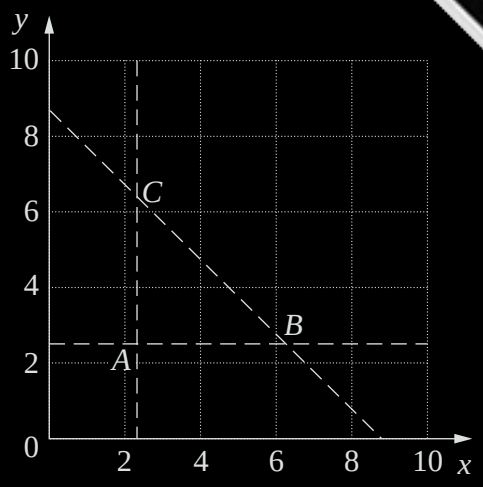
(i) both balls are green,

Answer (*b*)(i) .....[1]

(ii) the two balls have different colours.

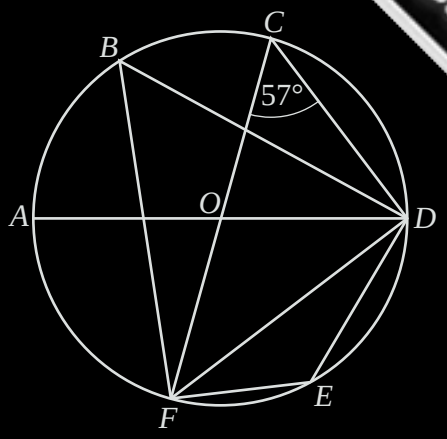
Answer (*b*)(ii) .....[1]

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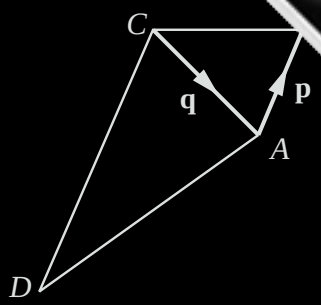


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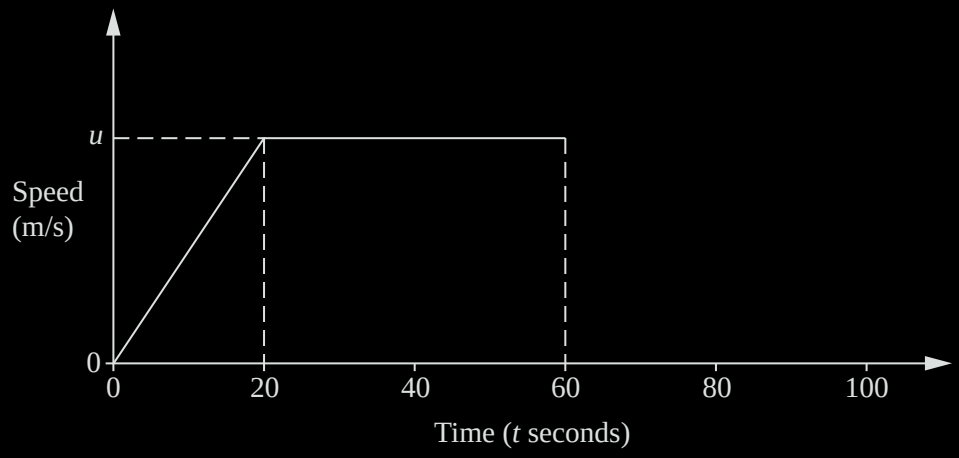
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The diagram is the speed-time graph of part of the journey of a car.
From $t = 0$ to $t = 20$ the car moves with a constant acceleration.
From $t = 20$ to $t = 60$ the car moves with a constant speed of u metres per second.

(a) When $t = 20$ the car has travelled D metres from the start.

Calculate the value of t when the car has travelled $2D$ metres **from the start**.

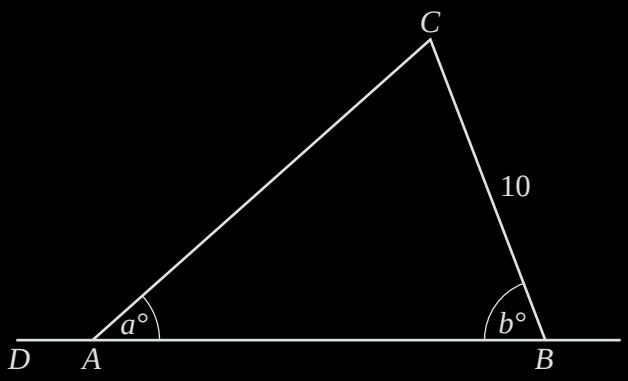
Answer (*a*) $t =$ .....[2]

(b) At $t = 60$, the car slows down with a constant deceleration.
This deceleration is half of the acceleration between $t = 0$ and $t = 20$.

During this period of deceleration, calculate the value of t when the car has a speed of $\frac{u}{4}$ metres per second.

Answer (*b*) $t =$ .....[2]

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$x°$	$a°$	$b°$
$\sin x°$	$\frac{3}{5}$	$\frac{24}{25}$
$\cos x°$	$\frac{4}{5}$	$\frac{7}{25}$
$\tan x°$	$\frac{3}{4}$	$\frac{24}{7}$

In the diagram, DAB is a straight line.
$BC = 10$ cm, $C\hat{A}B = a°$ and $C\hat{B}A = b°$.

Use as much information given in the table as is necessary to answer the following questions.

(a) Write down the value of $\cos D\hat{A}C$.

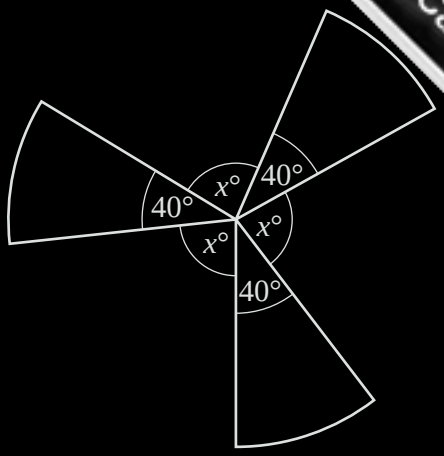
Answer (*a*) .....[1]

(b) Calculate AC.

Answer (*b*) $AC =$ ..... cm [3]

Question 27 is printed on the following page

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